

Matchbox Capacitor Tester

KAUSHIK HAZARIKA

Checking of an electrolytic capacitor without an LCR meter is not free from hassles. Here is a simple solution for checking the serviceability of a capacitor.

Circuit and working

Fig. 1 shows the circuit diagram of a simple capacitor tester which can be housed in a small matchbox. It is basically an oscillator circuit comprising two transistors, BC558 (T1) and BC548 (T2), and a few other components. The frequency of oscillation, which is in audio range, depends on value of the capacitor under test. The tester can check the serviceability of capacitors in the range of $1\mu\text{F}$ to $2200\mu\text{F}$.

Construction and testing

An actual-size, single-side PCB for the capacitor tester is shown in Fig. 2 and its component layout in Fig. 3.

Connector CON1 in the circuit is for inserting the capacitor under test. Four-pin berg strip female connector is used here to accommodate different sizes of capacitors. LS1 is a small speaker used as headphone (shown in Fig. 4). Assembled PCB of the circuit can be enclosed in a matchbox as shown in Fig. 5.

After wiring, do verify that voltages are as per Test Points table.

For testing the capacitor, insert it in CON1. If the capacitor is good, a

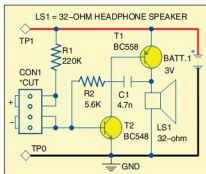


Fig. 1: Circuit diagram of the capacitor tester

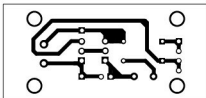


Fig. 2: An actual-size PCB layout for the tester circuit

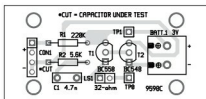


Fig. 3: Component layout of the PCB

Test Points

Test point	Details
TP0	0V
TP1	3V

fire-siren like sound is produced. A continuous beep is produced if the



Fig. 4: Earpiece of a headphone



Fig. 5: Capacitor tester in matchbox

PARTS LIST

Semiconductors:

- T1 - BC558 pnp transistor
- T2 - BC548 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 220-kilo-ohm
- R2 - 5.6-kilo-ohm

Capacitor:

- C1 - 4.7nF ceramic disk

Miscellaneous:

- LS1 - 32-ohm headphone speaker
- CON1 - 4-pin connector
- BATT.1 - 3V button cell
- 2-pin terminal connector

capacitor is short. ●

The author is a regular contributor to EFY

THE COMPLETE MAGAZINE ON OPEN SOURCE

OpenSource

ForYou



Your favourite Magazine on
Open Source is now on the Web, too.

OpenSourceForU.com

Follow us on Twitter@LinuxForYou